CAPSTONE PROJECT III

Bank Marketing Campaign

Predictive Dial ML for Term Deposit Subscription

Optimizing marketing efficiency through machine learning to identify high-potential term deposit subscribers and reduce campaign costs.



PRESENTED TO
Marketing Director

DATASET
7,813 customer records

MODEL
Random Forest

PRIMARY METRIC
Recall

OVERVIEW

Presentation Flow

01 **Business Problem**
Term deposit subscription prediction and cost-benefit framing

02 **Data Overview**
7,813 customer records · 10 features + 1 target from bank campaigns

03 **Exploratory Data Analysis**
Key patterns in customer behavior driving deposit decisions

04 **Data Preprocessing**
Simple Imputer (median), feature engineering, encoding and scaling pipeline

05 **Model Selection**
Cross-validated comparison of 8 ML algorithms via Recall

06 **Hyperparameter Tuning**
Random Forest optimization + threshold calibration at 0.40

07 **Model Evaluation**
Recall 75%, Precision 65% · confusion matrix deep-dive

08 **Business Impact**
39.4% cost reduction · $6,798/day saved vs. call-all

09 **SHAP Insights**
Feature importance and key business drivers explained

10 **Conclusion & Recommendations**
Tiered outreach strategy and deployment roadmap

BACKGROUND

Business Problem

Bank of America conducts telemarketing campaigns to sell term deposit products. The marketing team calls all customers **indiscriminately**, resulting in:

01 High operational costs from unnecessary calls

02 Agent fatigue, ineffective calls and customer frustration

03 No prioritization of high-potential leads

> ***The challenge:*** *build a predictive model that identifies* ***which*** *customers are most likely to subscribe to a term deposit, enabling the bank to focus resources on high-probability leads.*

DATASET AT A GLANCE · BALANCED

7,813
Customer records

10 + 1
Features + target column

48%
Positive class (Deposit = Yes)

BACKGROUND

Cost-Benefit Framework

	PREDICTED: NO	PREDICTED: YES
ACTUAL: NO	TN **True Negative** Correctly skips a non-depositor → cost saved	FP **False Positive** Calls a non-depositor → $0.10 wasted call cost
ACTUAL: YES	FN · HIGH COST **False Negative** Misses a depositor → $1.00 opportunity loss	TP **True Positive** Correctly identifies a depositor → revenue captured

STRATEGIC PRIORITY

Maximize Recall

to minimize missed depositors. A missed depositor costs 10× more than an unnecessary call.

FN cost basis: average annual Net Interest Margin of $300, normalized daily. Reference: FDIC Quarterly Banking Profile (Q4 2025), an official U.S. government report on the financial performance of 4,336 federally insured banks.

DATA UNDERSTANDING

Dataset Overview

Feature	Type	Description
age	Numeric	Age of the customer
job	Categorical	Type of job (admin, technician, services, etc.)
balance	Numeric	Average yearly account balance (EUR)
housing	Binary	Has a housing loan? (yes/no)
loan	Binary	Has a personal loan? (yes/no)
contact	Categorical	Contact type (cellular, telephone, unknown)
month	Categorical	Month of last contact
campaign	Numeric	Number of contacts during current campaign
pdays	Numeric	Days since last contact from previous campaign (−1 = not contacted)
poutcome	Categorical	Outcome of previous marketing campaign
deposit	Target	Did the client subscribe to a term deposit?

CLASS DISTRIBUTION



Yes · 3,732 (47.8%) **No** · 4,081 (52.2%)

TRAIN / TEST SPLIT



80% train · 6,250 **20%** test · 1,563

Stratified split preserves the 48/52 class ratio in both sets.

No data leakage — preprocessing fitted inside each CV fold only.

EXPLORATORY DATA ANALYSIS · 1 OF 2

Key EDA Findings

01 ↑ Positive signal

Prior Engagement Drives Conversion

Customers never contacted before show the highest non-deposit rate. Customers with longer prior engagement are significantly more likely to convert.

INSIGHT

Shorter engagement history = lower commitment to new products.

02 ↑ Positive signal

Balance Matters

Customers with higher average yearly balances show stronger propensity to deposit. Outliers exist but are valid business values — retained for modeling.

INSIGHT

Wealth segmentation should inform priority tier assignments.

03 ↓ Negative signal

Contact Type Is a Predictor

Customers with an ‘unknown’ contact type are far less likely to deposit. Cellular contact performs best.

INSIGHT

Verified contact channels are critical for campaign effectiveness.

Key EDA Findings

04 ◆ Timing signal

Month of Contact Is Significant

May, June, July and August see higher campaign activity but variable success. March, September, October and December show higher conversion rates.

INSIGHT

Campaign timing should align with historically successful months.

05 ↑ Positive signal

Previous Campaign Success = Best Predictor

Customers who converted in a previous campaign are far more likely to convert again — the strongest positive signal in the data.

INSIGHT

Customer retention and repeat engagement programs are high-ROI.

06 ↓ Negative signal

Housing Loan = Negative Indicator

Customers with active housing loans are less likely to deposit — likely due to limited disposable income and existing bank commitments.

INSIGHT

Deprioritize housing-loan customers in outreach tiers.

DATA TRANSFORMATION

Preprocessing Pipeline

1 **Feature Engineering**

Encode target: yes → 1, no → 0 · Create **was_contacted_before** binary flag from pdays · Replace pdays = −1 with 0

2 **Column Transformer**

SimpleImputer (median) + RobustScaler → numeric · OrdinalEncoder → housing, loan · OneHotEncoder → job, contact, month, poutcome

3 **Train-Test Split**

80/20 split · stratify=y to preserve class distribution · random_state=0 for reproducibility

4 **Pipeline Wrapper**

Preprocessing inside Pipeline · Fit only on training folds — prevents data leakage in CV · Transform applied uniformly

MULTICOLLINEARITY FINDING

Spearman correlation shows high correlation between **was_contacted_before** and **pdays**. Both were intentionally retained: ensemble methods handle multicollinearity via feature diversity at each split, and both features carry distinct predictive signal.

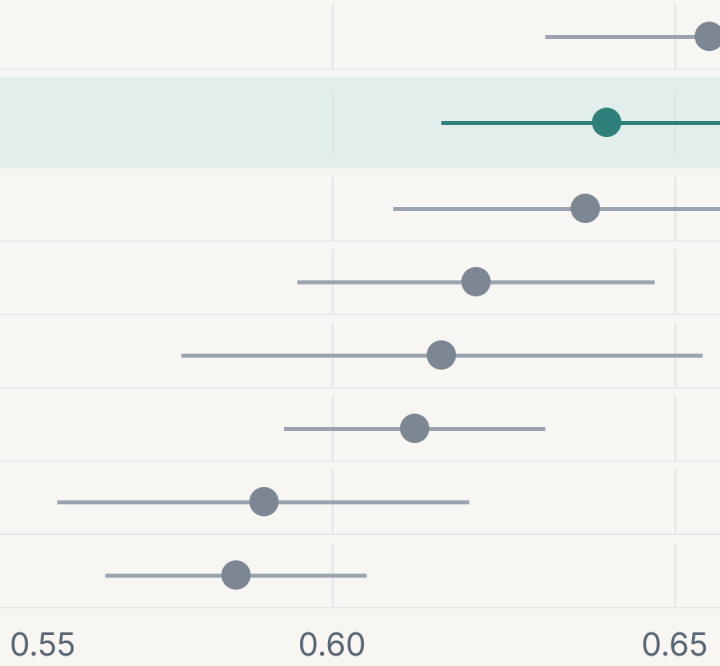
WHY ROBUSTSCALER?

balance and **campaign** contain significant outliers that are valid business extremes. RobustScaler uses median + IQR instead of mean + std — resistant to extreme values without discarding them.

MODELING

Model Selection — 5-Fold Cross-Validation (Recall)

Model	Type	Recall mean ± std (5-fold CV)	Mean	Std	
AdaBoost	Ensemble / Boost		0.655	±0.024	
Random Forest	Ensemble / Bag		**0.640**	±0.024	SELECTED
Voting	Ensemble / Soft		0.637	±0.028	
KNN	Individual		0.621	±0.026	
Stacking	Ensemble / Meta		0.616	±0.038	
Decision Tree	Individual		0.612	±0.019	
Logistic Reg.	Individual		0.590	±0.030	
Bagging	Ensemble / Bag		0.586	±0.019	



MODELING

Why Random Forest?

CV RECALL — TOP TWO

Random Forest · selected

0.640

± 0.024

AdaBoost

0.655

± 0.024

AdaBoost scored slightly higher on mean Recall; Random Forest was chosen for the strategic reasons at right.

1 **Handles multicollinearity**

Feature diversity at each split neutralizes correlated features naturally.

2 **Robust to outliers**

Averaging (bagging) reduces the impact of extreme values in balance and campaign.

3 **Overfitting control**

AdaBoost is more susceptible to noise; Random Forest's averaging provides stability.

4 **Class imbalance**

class_weight='balanced' handles the minority class effectively.

5 **SHAP compatibility**

Tree-based SHAP explainer provides reliable business interpretability.

MODELING

Hyperparameter Tuning

RANDOMIZEDSEARCHCV · 50 ITERATIONS × 5-FOLD CV

Parameter	Search space	Best	Effect
n_estimators	[100, 200, 300, 500]	300	More trees → lower variance
max_depth	[5, 8, 10, 15, None]	None	Shallower → less overfitting
min_samples_leaf	[1, 5, 10, 20]	1	Larger → smoother boundaries
min_samples_split	[2, 5, 10, 20]	10	Fewer splits → less overfit
max_features	['sqrt', 'log2', 0.5]	log2	Diversity between trees
class_weight	['balanced', None]	balanced	Handles near-imbalance

BEST CV RECALL

0.65

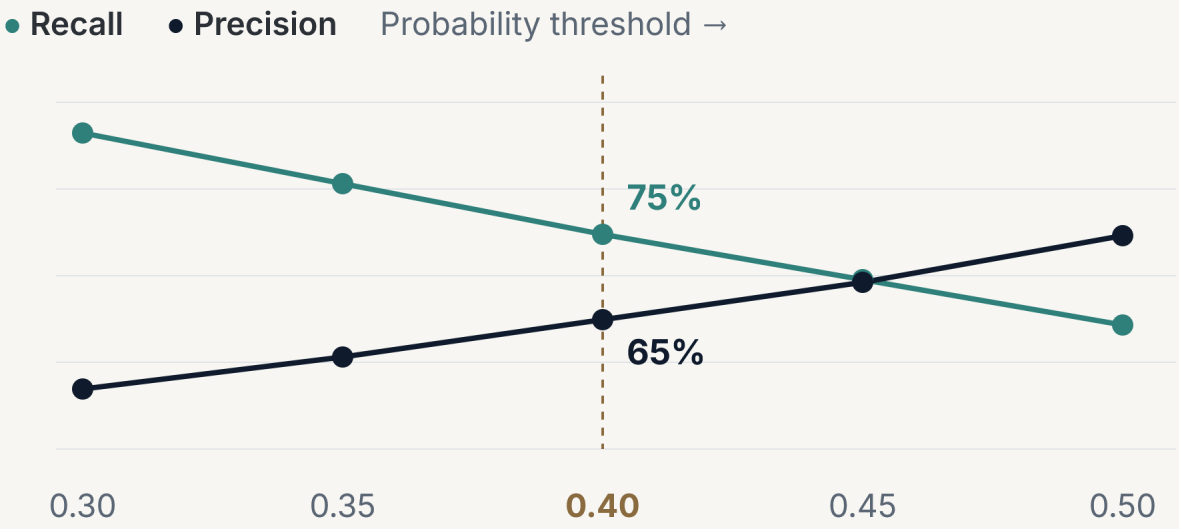
RECALL IMPROVEMENT AFTER TUNING · TEST SET

62% Base model → **75%** Tuned · threshold 0.40

+12.6 pp Recall, overfit gap cut from 0.38 to 0.20

MODELING

Threshold Calibration — Why 0.40?



Threshold	Recall	Precision	Strategy
0.30	86%	57%	Aggressive
0.35	81%	61%	High Recall
0.40	75%	65%	★ Optimal
0.45	69%	69%	Balanced
0.50	64%	75%	Conservative

Recall = 75%
Captures 3 out of 4 actual depositors

Precision = 65%
Majority of contacted leads are genuine prospects

~4,296 calls / day
Down from 7,813 — 45% fewer calls

$6,798 / day
Net cost savings vs. calling all customers

MODELING

Model Evaluation — Final Performance

RECALL · PRIMARY KPI
75%
75 of every 100 actual depositors captured

PRECISION
65%
65% of predicted depositors actually convert

F1-SCORE
0.69
Harmonic mean of Recall & Precision

ACCURACY
69%
Correct classifications on the test set

	PREDICTED: NO	PREDICTED: YES
ACTUAL: NO	TN **513** Correctly identified non-depositors	FP **303** Called but did not convert
ACTUAL: YES	FN **188** Missed actual depositors	TP **559** Correctly identified depositors ✓

CLASSIFICATION REPORT · TEST SET

Class	Precision	Recall	F1
No Deposit (0)	0.73	0.63	0.68
Deposit (1)	0.65	0.75	0.69

OVERFITTING CHECK · RECALL



Gap = 0.20 — generalizes reasonably well; class_weight='balanced' reduces overfit tendency.

BUSINESS IMPACT

Model vs. No Model — Daily Economics

$6,798
saved per day

39.4%
total cost reduction

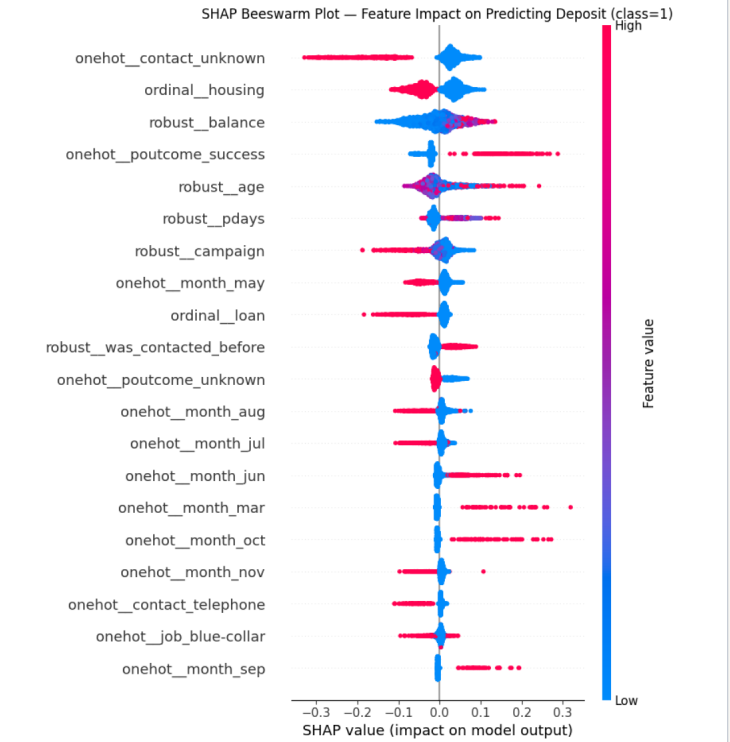
39
staff reallocated

3,517
fewer calls per day

Metric	Call all	Basis	With model	Basis
Total calls / day	7,813	All leads contacted	4,296	(3,732 × 75%) ÷ 65%
Depositors captured	3,732	100% Recall	2,793	75% Recall maintained
Missed opportunities	0	No false negatives	939	3,732 − 2,793
Operational call cost	$781/day	$0.10 × 7,813 calls	$430/day	$0.10 × 4,296 calls
Opportunity loss	$0/day	All depositors contacted	$939/day	939 FN × $1
Staff required	87 agents	4 min/call, 6 productive hr/day	48 agents	39 staff freed up
Staff overhead	$16,477/day	$362,500 / month	$9,091/day	$200,000 / month
Total daily cost	$17,258/day	High operational burden	$10,460/day	39.4% lower

EXPLAINABILITY

SHAP Beeswarm — Top 5 Drivers Explained



Red = high feature value · Blue = low

1 ↓ **Unknown contact channel**

If the contact method is unknown (red), it significantly decreases the chance of a deposit. Known contact methods are crucial.

2 ↓ **Housing loan = negative**

Customers with a housing loan (red) are less likely to subscribe, likely due to existing financial commitments.

3 ↑ **Account balance**

Higher account balances (red) generally lead to a higher probability of subscribing.

4 ↑ **Previous campaign success**

If the previous campaign was a success (red), there is a very strong positive impact on the prediction.

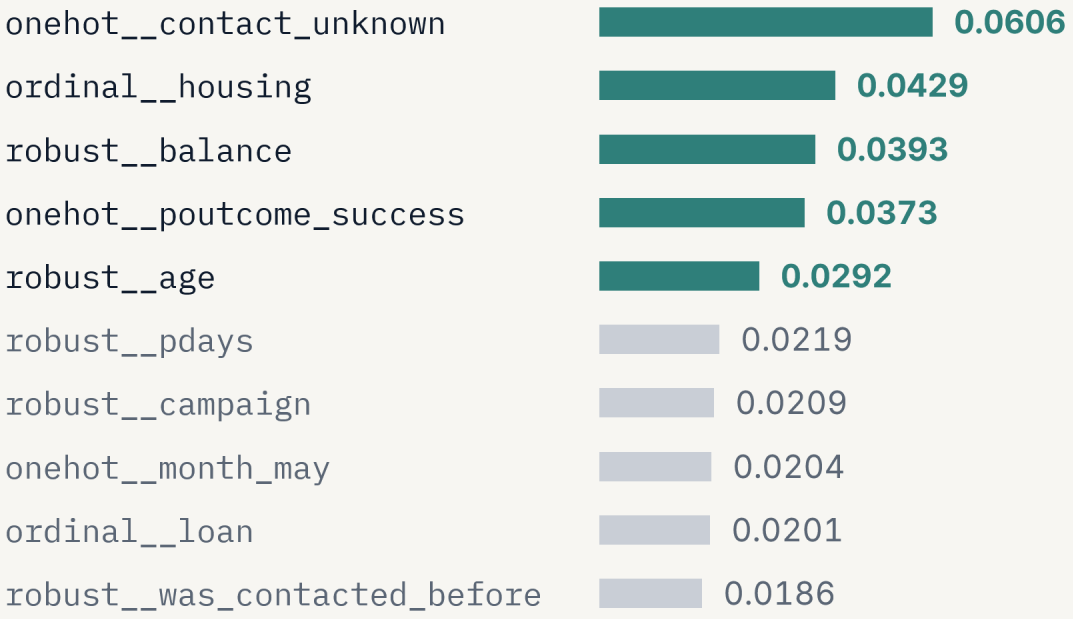
5 ↑ **Age**

Older customers (red) show a trend of being more likely to deposit than younger ones.

EXPLAINABILITY

Feature Importance → Business Action

TOP 10 · MEAN |SHAP| VALUE



BUSINESS INTERPRETATION

1. **Clean the CRM.** Unknown contact type is the No. 1 predictor — verify contact data before campaigns.
2. **Deprioritize housing-loan customers** or segment this group separately in outreach planning.
3. **Segment by wealth.** Higher-balance customers convert more — consider premium tiering.
4. **Prioritize repeat customers.** Prior converters are significantly more likely to convert again.
5. **Target older customers,** who are more likely to subscribe to the term deposit program.

CONCLUSION & RECOMMENDATIONS

Summary of Results

75%

RECALL

75 of every 100 depositors successfully identified

65%

PRECISION

Majority of leads contacted are genuine prospects

39.4%

COST REDUCTION

$6,798 saved daily vs. call-all strategy

39

STAFF FREED

High-value reallocation to other revenue streams

Tuned Random Forest · Threshold 0.40 · Recall 75% · Precision 65% · $6,798/day savings

CONCLUSION & RECOMMENDATIONS

Strategic Recommendations

A Tiered Outreach Strategy

Segment leads by predict_proba score:

Priority 1	Priority 2	Priority 3
> 0.80	0.40 – 0.80	< 0.40
Senior agents, same-day calls	Standard campaign queue	Email/SMS, low-cost digital nurturing

B CRM Data Quality Program

Unknown contact type is a major negative predictor. Invest in contact verification (KYC) so cellular and telephone channels are validated before launch.

C Seasonal Campaign Scheduling

Focus high-intensity campaigns in March, September, October and December — historically highest-conversion months (EDA & SHAP); reduce intensity in May.

D Retention & Repeat Engagement

Previous campaign success is the strongest positive predictor. Build a loyal-customer re-engagement program that targets prior subscribers first.

From Insight to Prioritized Leads

PHASE 1 SHAP Insights & Key Drivers

↓ contact_unknown (Unknown Contact)

Strongest negative predictor. High unknown values reduce success probability.

↓ housing (Housing Loan)

Customers with existing housing loans have financial commitments and are less likely to deposit.

↑ poutcome_success (Previous Success)

A strong positive predictor. Customers with previous success history are highly valuable.

Key negative and positive drivers identified

PHASE 2 Lead Segmentation & Prioritization

Priority 1 · High certainty (prob > 0.80)

Senior Relationship Manager

Focus on high-balance and poutcome_success leads.

Priority 2 · Standard target (prob 0.40–0.80)

Standard Campaign Script

Focus on older age groups and no-loan leads.

Priority 3 · Low probability (prob < 0.40)

Low-Cost Automation

Leads with contact_unknown or high call frequency (campaign) are deprioritized.

Leads tiered by predict_proba score

Pilot Execution

PHASE 3 Pilot Execution (30–90 days)

Dynamic Thresholding

Maintain threshold at 0.40 — captures 75% of potential depositors at optimized cost. Recalibrate if liquidity needs, interest rates or labor costs shift.

Marketing Optimization

Develop “Returning Customer” loyalty offers. Align deposit offers with the debt cycles of customers holding existing loans.

Data Integrity & KYC

Improve Know Your Customer (KYC) so valid telephone and cellular contact data is captured before outreach.

Operational Focus & Call Caps

Shift KPIs from call volume to engagement quality. Prevent customer fatigue with call-frequency limits.

$6,798 daily value generated from model use

Monitoring, ROI & Governance

PHASE 4 Monitoring & ROI

75%
Recall at 0.40 threshold
Captures 3/4 of potential depositors

39.4%
Reduction in daily economic burden
From $17,258 to $10,460 per day

3,517
Operational waste reduction
Unnecessary calls eliminated daily

$6,798 daily economic value created

PHASE 5 Governance

Monitor for feature drift

Performance review every 3–6 months; macroeconomic shifts (e.g., interest rates) considered; model governance protocols followed.

Recall holding at benchmark (≥ 70%)?

YES
Monitor & Deploy
Reallocate resources to other high-value revenue streams.

NO
Retrain Model
Ingest fresh data, update features, recalibrate parameters.

Cost-Optimization Engine active